week8_notebook_student

November 12, 2026

1 HS4002 Week 8

1.1 Practising with Count Models

Research question: **what factors influence the number of children someone has?**

We'll compare three model types for count outcomes: 1. OLS (baseline) 2. Poisson regression 3. Negative Binomial regression

Plus: overdispersion testing and marginal effects.

[1]:
```
# !pip install pandas numpy plotnine statsmodels stargazer marginaleffects scipy

import pandas as pd
import numpy as np
from plotnine import *
import warnings
warnings.filterwarnings('ignore')
import statsmodels.formula.api as smf
from scipy.stats import chi2
```

[2]:
```
raw_df = pd.read_csv("gss2022_mini2.csv")

chosen_variables = [
        'id', 'age', 'happy', 'yrlvmus', 'yrartxbt', 'yrmovie', 'yrcreat',
        'income16', 'prestg10', 'degree', 'race', 'sex', 'wrkstat', 'biokids',␣
 ↪'educ'
]
df = raw_df[chosen_variables].dropna().copy()

income_map = {
        1: 500,    2: 2000,   3: 3500,   4: 4500,   5: 5500,   6: 6500,
        7: 7500,   8: 9000,   9: 11250,  10: 13750, 11: 16250, 12: 18750,
        13: 21250, 14: 23750, 15: 27500, 16: 32500, 17: 37500, 18: 45000,
        19: 55000, 20: 67500, 21: 82500, 22: 100000, 23: 120000, 24: 140000,
        25: 160000, 26: 250000
}
df['income_cont']   = df['income16'].map(income_map)
df['income_10k']   = df['income_cont'] / 10000
```

```
df['age']       = raw_df.loc[df.index, 'age'] - 18

df['binary_lvmus']  = (df['yrlvmus']  == 1).astype(int)
df['binary_artxbt'] = (df['yrartxbt'] == 1).astype(int)
df['binary_movie']  = (df['yrmovie']  == 1).astype(int)
df['binary_creat']  = (df['yrcreat']  == 1).astype(int)
df['omni'] = df[['binary_lvmus','binary_artxbt','binary_movie','binary_creat']].
 ↪sum(axis=1)

df['race_bin']   = pd.Categorical(df['race'].astype(str))
df['race_cat']   = df['race'].map({1: 'White', 2: 'Black'}).fillna('Other')
df['ba_binary']  = (df['degree'] >= 3).astype(int)
df['sex_woman']  = pd.Categorical(
    np.where(df['sex'] == 2, 'Woman', 'Not Woman'),
    categories=['Not Woman', 'Woman']
)
df['working']    = np.where(df['wrkstat'] <= 3, 'Working', 'Not Working')
df['happy_bin']  = (df['happy'] <= 2).astype(int)
df['happy_cat']  = np.where(df['happy'] <= 2, 'Happy', 'Not Happy')
```

2 Starting with an OLS model

Regress no. of children (`biokids`) against income in ten thousands (`income_10k`). Note: one unit = $10,000. Name this `ols_lm1`.

[3]:
```
ols_lm1 = smf.ols('biokids ~ income_10k', data=df).fit()
print(ols_lm1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                biokids   R-squared:                       0.000
Model:                            OLS   Adj. R-squared:                 -0.002
Method:                 Least Squares   F-statistic:                   0.04228
Date:                Thu, 12 Nov 2026   Prob (F-statistic):              0.837
Time:                        01:06:25   Log-Likelihood:                -1050.5
No. Observations:                 557   AIC:                             2105.
Df Residuals:                     555   BIC:                             2114.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      1.5647      0.107     14.639      0.000       1.355       1.775
income_10k     0.0020      0.010      0.206      0.837      -0.017       0.021
==============================================================================
Omnibus:                      147.242   Durbin-Watson:                   1.830
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              411.395
```

```
Skew:                          1.287   Prob(JB):                     4.64e-90
Kurtosis:                      6.332   Cond. No.                         17.7
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

How would you interpret the coefficient for `income_10k`? What is unusual about using OLS on a count variable? an increase in income by 10k is associated with an increase of 0.0020 children. unusual to use OLS as you can't have 0.0020 children

3 Continuing with a Poisson Model

Use `smf.poisson()` — equivalent to `glm(..., family='poisson')` in R.

[ ]:
```
# poisson model assumes that most people have the same number of kids
po_lm1 = smf.poisson('biokids ~ income_10k', data=df).fit()
print(po_lm1.summary())
```

```
Optimization terminated successfully.
         Current function value: 1.767601
         Iterations 3
                          Poisson Regression Results
==============================================================================
Dep. Variable:                 biokids   No. Observations:                  557
Model:                         Poisson   Df Residuals:                      555
Method:                            MLE   Df Model:                            1
Date:                Thu, 12 Nov 2026   Pseudo R-squ.:               3.454e-05
Time:                        01:06:54   Log-Likelihood:                -984.55
converged:                       True   LL-Null:                       -984.59
Covariance Type:            nonrobust   LLR p-value:                    0.7942
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.4478      0.053      8.410      0.000       0.343       0.552
income_10k     0.0012      0.005      0.261      0.794      -0.008       0.011
==============================================================================
```

Interpretation: a 10k increase in income is expected to cause a 0.0012 increase in the log count of number of children, holding all else constant

[ ]:
```
import numpy as np
b = po_lm1.params['income_10k']
print(f'Factor of change, exp(b): {np.exp(b):.4f}')
print(f'Percentage change, 100 * [exp(b) - 1]: {100 * (np.exp(b) - 1):.2f}%')
```

```
Factor of change, exp(b): 1.0012
Percentage change, 100 * [exp(b) - 1]: 0.12%
```

4 Adding Age

Add `age` to both models. `age` is centered at 18, so 0 is an 18-year-old. Save as `ols_lm2` and `po_lm2`.

[6]:
```
ols_lm2 = smf.ols('biokids ~ income_10k + age', data=df).fit()
po_lm2  = smf.poisson('biokids ~ income_10k + age', data=df).fit()
print(po_lm2.summary())
```

```
Optimization terminated successfully.
         Current function value: 1.720780
         Iterations 5
                          Poisson Regression Results
==============================================================================
Dep. Variable:                 biokids   No. Observations:                  557
Model:                         Poisson   Df Residuals:                      554
Method:                            MLE   Df Model:                            2
Date:                Thu, 12 Nov 2026   Pseudo R-squ.:                 0.02652
Time:                        01:11:04   Log-Likelihood:                -958.47
converged:                       True   LL-Null:                       -984.59
Covariance Type:            nonrobust   LLR p-value:                 4.563e-12
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -0.0674      0.094     -0.720      0.472      -0.251       0.116
income_10k     0.0008      0.005      0.165      0.869      -0.009       0.010
age            0.0143      0.002      7.151      0.000       0.010       0.018
==============================================================================
```

4.1 Comparing the Poisson Models via Likelihood Ratio Test

[ ]:
```
lr_stat = 2 * (po_lm2.llf - po_lm1.llf)
df_diff = po_lm2.df_model - po_lm1.df_model
p_val   = chi2.sf(lr_stat, df_diff)
print(f'LR statistic = {lr_stat:.4f}, df = {df_diff:.0f}, p-value = {p_val:.
↪6f}')
# since p-value is < 0.01, we reject the null hypothesis. adding the variabl␣
↪age made the model fit better
```

```
LR statistic = 52.1579, df = 1, p-value = 0.000000
```

5 Building a Negative Binomial Model

Negative binomial relaxes the Poisson assumption that variance = mean (allowing overdispersion).

Equivalent to R's `MASS::glm.nb()`.

Read the coefficients the same way as in a Poisson model: as a factor of change, exp( ), or as a percentage change, $100 \times [\exp(\ ) - 1]\%$.

[ ]:
```
# negative binomial model allows for messier counts --> people have variosu␣
↪number of kids and values are spread out

nb_m1 = smf.negativebinomial('biokids ~ income_10k + age', data=df).fit()
print(nb_m1.summary())
```

```
Optimization terminated successfully.
         Current function value: 1.670251
         Iterations: 14
         Function evaluations: 19
         Gradient evaluations: 19
                     NegativeBinomial Regression Results
==============================================================================
Dep. Variable:                biokids   No. Observations:                  557
Model:               NegativeBinomial   Df Residuals:                      554
Method:                           MLE   Df Model:                            2
Date:                Thu, 12 Nov 2026   Pseudo R-squ.:                 0.01718
Time:                        01:20:09   Log-Likelihood:                -930.33
converged:                       True   LL-Null:                       -946.59
Covariance Type:            nonrobust   LLR p-value:                 8.681e-08
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -0.0832      0.116     -0.718      0.473      -0.310       0.144
income_10k     0.0005      0.006      0.081      0.936      -0.011       0.012
age            0.0148      0.003      5.755      0.000       0.010       0.020
alpha          0.3822      0.073      5.212      0.000       0.239       0.526
==============================================================================
```

interpretation of alpha: since alpha > 0 (p <0.01), the counts are messy. using negative binomial is a good idea

[9]:
```
from stargazer.stargazer import Stargazer

sg = Stargazer([ols_lm2, po_lm2, nb_m1])
sg.title('Number of Children')
sg.custom_columns(['OLS', 'Poisson', 'Neg. Binomial'], [1, 1, 1])

# Ending the cell on the object itself renders the table in the notebook
sg
```

[9]: <stargazer.stargazer.Stargazer at 0x146f52ea0>

5.1 Test for Overdispersion

Was NB necessary? We test whether the NB's dispersion parameter (alpha) is significantly different from 0. If alpha = 0, the NB reduces to Poisson.

This is equivalent to R's `pscl::odTest()`.

[33]:
```
# Overdispersion test: run a likelihood ratio test!
p_od  = 0.5 * chi2.sf(lr_od, 1) # looks up the p-value of alpha
print(f'Overdispersion LR statistic = {lr_od:.4f}')
print(f'p-value (one-sided) = {p_od:.6f}')
print('NB preferred over Poisson:', p_od < 0.05)
# # since p-value is < 0.01, we reject the null hypothesis. negative binomial␣
 ↪model is better
```

```
Overdispersion LR statistic = 56.2895
p-value (one-sided) = 0.000000
NB preferred over Poisson: True
```

6 Marginal Effects

A marginal effect is the expected change in the number of children when an independent variable increases by one unit, ceteris paribus. Compute them the same way as for the logit models.

[11]:
```
from marginaleffects import avg_comparisons, comparisons
```

6.1 Average marginal effects of income

[12]:
```
avg_comparisons(po_lm2, variables='income_10k')
```

[12]:
```
shape: (1, 7)

 Estimate   Std.Error   z       P(>|z|)   S       2.5%      97.5%
 ---        ---         ---     ---       ---     ---       ---
 str        str         str     str       str     str       str

 0.00124    0.0075      0.165   0.869     0.203   -0.0135   0.016

Term: income_10k
Contrast: +1
```

a 10k increase of income is associated, on average across the sample, with an increase of 0.001238 in the number of children, holding all else constant. at p = 0.868, we do not reject the null hypothesis. there is no effect at the 0.05 threshold.

[13]:
```
avg_comparisons(nb_m1, variables='income_10k')
```

[13]:
```
shape: (1, 7)

 Estimate   Std.Error   z       P(>|z|)   S       2.5%      97.5%
 ---        ---         ---     ---       ---     ---       ---
 str        str         str     str       str     str       str

 0.00077    0.00953     0.0808  0.936     0.096   -0.018    0.0195
```

```
Term: income_10k
Contrast: +1
```

a 10k increase of income is associated, on average across the sample, with an increase of 0.00077 in the number of children, holding all else constant. at p = 0.0959, we do not reject the null hypotehsis. there is no effect

6.2 Marginal effect at the mean of income

[14]:
```
comparisons(po_lm2, variables='income_10k', newdata='mean')
```

[14]:
```
shape: (1, 7)

  Estimate   Std.Error   z       P(>|z|)   S       2.5%      97.5%
  ---        ---         ---     ---       ---     ---       ---
  str        str         str     str       str     str       str

  0.0012     0.00727     0.165   0.869     0.203   -0.0131   0.0155

Term: income_10k
Contrast: +1
```

a 10k increase in income is associated, for a typical average person, with an increase of 0.001201 number of children, holding all else constant. at p = 0.8688, we do not reject the null hypothesis. there is no effect at the 0.05 threshold

[15]:
```
comparisons(nb_m1, variables='income_10k', newdata='mean')
```

[15]:
```
shape: (1, 7)

  Estimate   Std.Error   z        P(>|z|)   S       2.5%      97.5%
  ---        ---         ---      ---       ---     ---       ---
  str        str         str      str       str     str       str

  0.000746   0.00923     0.0808   0.936     0.096   -0.0174   0.0189

Term: income_10k
Contrast: +1
```

a 10k increase in income is associated, for a typical average person, with an increase of 0.0000746 number of children, holding all else constant. at p = 0.935, we do not reject the null hypothesis. there is no effect at the 0.05 threshold

7 Independent Investigation: Education and Number of Children

Investigate the relationship between **no. of children** (`biokids`) and **BA education** (`ba_binary`).

Do the following:

1. Include these independent variables: `ba_binary`, `age`, `income_10k`, `sex_woman`, and `race_bin`.
2. Build OLS, Poisson, and Negative Binomial models.
3. Produce a Stargazer table comparing the three models.
4. Compute marginal effects for each model.
5. Decide which model is best. Justify.
6. Answer: does a person with a BA have more or fewer children, and by how much? For the Poisson and negative binomial models, report a factor of change or a percentage change.

[17]:
```
# OLS model
ols_lm2 = smf.ols('biokids ~ ba_binary + age + income_10k + sex_woman +␣
 ↪race_bin', data=df).fit()
print(ols_lm2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 biokids   R-squared:                       0.118
Model:                             OLS   Adj. R-squared:                  0.109
Method:                  Least Squares   F-statistic:                     12.29
Date:                 Thu, 12 Nov 2026   Prob (F-statistic):           5.36e-13
Time:                         01:41:55   Log-Likelihood:                -1015.5
No. Observations:                 557   AIC:                             2045.
Df Residuals:                     550   BIC:                             2075.
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.4837      0.190      2.547      0.011       0.111       0.857
sex_woman[T.Woman]     0.4935      0.130      3.798      0.000       0.238       0.749
race_bin[T.2.0]     0.6263      0.201      3.117      0.002       0.232       1.021
race_bin[T.3.0]     0.0643      0.282      0.228      0.820      -0.489       0.618
ba_binary     -0.4814      0.141     -3.410      0.001      -0.759      -0.204
age            0.0233      0.004      6.202      0.000       0.016       0.031
income_10k     0.0216      0.010      2.165      0.031       0.002       0.041
==============================================================================
Omnibus:                      156.400   Durbin-Watson:                   1.934
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              482.642
```

```
Skew:                          1.327   Prob(JB):                     1.57e-105
Kurtosis:                      6.709   Cond. No.                         175.
================================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

[18]:
```
# Poisson model
po_lm2 = smf.poisson('biokids ~ ba_binary + age + income_10k + sex_woman +␣
 ↪race_bin', data=df).fit()
print(po_lm2.summary())
```

```
Optimization terminated successfully.
         Current function value: 1.671346
         Iterations 5
                          Poisson Regression Results
==============================================================================
Dep. Variable:                biokids   No. Observations:                  557
Model:                        Poisson   Df Residuals:                      550
Method:                           MLE   Df Model:                            6
Date:                Thu, 12 Nov 2026   Pseudo R-squ.:                 0.05449
Time:                        01:42:33   Log-Likelihood:                -930.94
converged:                       True   LL-Null:                       -984.59
Covariance Type:            nonrobust   LLR p-value:                 7.505e-21
==============================================================================
======
                         coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
------
Intercept             -0.3234      0.112     -2.885      0.004      -0.543
-0.104
sex_woman[T.Woman]     0.3184      0.070      4.564      0.000       0.182
0.455
race_bin[T.2.0]        0.3711      0.094      3.955      0.000       0.187
0.555
race_bin[T.3.0]        0.0390      0.157      0.249      0.803      -0.268
0.346
ba_binary             -0.3114      0.075     -4.136      0.000      -0.459
-0.164
age                    0.0154      0.002      7.507      0.000       0.011
0.019
income_10k             0.0142      0.005      2.749      0.006       0.004
0.024
==============================================================================
======
```

Interpretation: having a ba education is expected to cause a 0.3114 ($p < 0.01$) decrease in the log count of number of children, holding all else constant

[ ]:
```
# negative binomial model
nb_m2 = smf.negativebinomial('biokids ~ ba_binary + age + income_10k +␣
 ↪sex_woman + race_bin', data=df).fit()
print(nb_m2.summary())
```

```
Optimization terminated successfully.
         Current function value: 1.637030
         Iterations: 28
         Function evaluations: 34
         Gradient evaluations: 34
                     NegativeBinomial Regression Results
==============================================================================
Dep. Variable:                biokids   No. Observations:                  557
Model:               NegativeBinomial   Df Residuals:                      550
Method:                           MLE   Df Model:                            6
Date:                Thu, 12 Nov 2026   Pseudo R-squ.:                 0.03673
Time:                        01:43:06   Log-Likelihood:                -911.83
converged:                       True   LL-Null:                       -946.59
Covariance Type:            nonrobust   LLR p-value:                 5.111e-13
==============================================================================
======
                           coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
------
Intercept             -0.3463      0.134     -2.591      0.010      -0.608
-0.084
sex_woman[T.Woman]     0.3242      0.085      3.808      0.000       0.157
0.491
race_bin[T.2.0]        0.3756      0.119      3.143      0.002       0.141
0.610
race_bin[T.3.0]        0.0414      0.189      0.219      0.827      -0.330
0.413
ba_binary             -0.3301      0.093     -3.551      0.000      -0.512
-0.148
age                    0.0160      0.003      6.348      0.000       0.011
0.021
income_10k             0.0149      0.006      2.322      0.020       0.002
0.028
alpha                  0.2962      0.066      4.486      0.000       0.167
0.426
==============================================================================
======
```

Interpretation: having a ba education is expected to cause a 0.3301 decrease in the log count of

number of children ($p < 0.01$), holding all else constant

[34]:
```
# Overdispersion test
lr_od2 = 2 * (nb_m2.llf - po_lm2.llf)
p_od2  = 0.5 * chi2.sf(lr_od2, 1) # looks up the p-value of alpha
print(f'Overdispersion LR statistic = {lr_od2:.4f}')
print(f'p-value (one-sided) = {p_od2:.6f}')
print('NB preferred over Poisson:', p_od2 < 0.05)
```

```
Overdispersion LR statistic = 38.2281
p-value (one-sided) = 0.000000
NB preferred over Poisson: True
```

[26]:
```
# compare models using staragazer

from stargazer.stargazer import Stargazer

sg = Stargazer([ols_lm2, po_lm2, nb_m2])
sg.title('Sex, Race, Education, Age, Income')

sg.dependent_variable_name('Biokids')
sg.custom_columns(['Model 1', 'Model 2', 'Model 3'], [1, 1, 1])
sg.covariate_order(['ba_binary', 'age', 'income_10k', 'sex_woman[T.Woman]',␣
 ↪'race_bin[T.2.0]', 'Intercept'])
sg.rename_covariates({
'ba_binary': 'BA education',
'age': 'Age',
'income_10k': 'Income',
'sex_woman[T.Woman]': 'Sex',
'race_bin[T.2.0]': 'Race',
'Intercept': 'Intercept'
})

sg
```

[26]:
```
<stargazer.stargazer.Stargazer at 0x16a3589b0>
```

[ ]:
```
# average marginal effects for OLS model
avg_comparisons(ols_lm2, variables='ba_binary')
```

[ ]:
```
shape: (1, 7)

 Estimate   Std.Error   z       P(>|z|)    S      2.5%     97.5%
 ---        ---         ---     ---        ---    ---      ---
 str        str         str     str        str    str      str

 -0.515     0.145       -3.55   0.000419   11.2   -0.801   -0.23
```

```
Term: ba_binary
Contrast: 1 - 0
```

[28]:
```
# average marginal effect for Poisson
avg_comparisons(po_lm2, variables='ba_binary')
```

[28]:
```
shape: (1, 7)

 Estimate  Std.Error  z      P(>|z|)   S     2.5%    97.5%
 ---       ---        ---    ---       ---   ---     ---
 str       str        str    str       str   str     str

 -0.486    0.117      -4.17  3.55e-05  14.8  -0.715  -0.257

Term: ba_binary
Contrast: 1 - 0
```

[29]:
```
# average marginal effect for NB
avg_comparisons(nb_m2, variables='ba_binary')
```

[29]:
```
shape: (1, 7)

 Estimate  Std.Error  z      P(>|z|)   S     2.5%    97.5%
 ---       ---        ---    ---       ---   ---     ---
 str       str        str    str       str   str     str

 -0.515    0.145      -3.55  0.000419  11.2  -0.801  -0.23

Term: ba_binary
Contrast: 1 - 0
```

[35]:
```
# comparing aic values
print(f'AIC ols_lm2: {ols_lm2.aic:.2f}')
print(f'AIC po_lm2: {po_lm2.aic:.2f}')
print(f'AIC nb_m2: {nb_m2.aic:.2f}')
```

```
AIC ols_lm2: 2044.94
AIC po_lm2: 1875.88
AIC nb_m2: 1839.65
```

the negative binomial model is preferred. firstly, it has the lowest AIC value among all 3 models. secondly, after running a likelihood ratio test, we can see that the data is quite spread out, and thus a negative binomial model is more appropriate to use than a poisson model. lastly, after running average marginal effect tests for each model, we can see that the p-value for the negative binomial model is < 0.001, so we reject the null hypotehsis that there education has no effect on the number of children